

A Appendix

A.1 GLOW Algorithm

Algorithm 1 GETPROMPT: Generate GLOW Prompt from Input OWA Question over KG

Require: Q the user input question, $GLOW_v$ The system variation .

```

1: function GETPROMPT( $Q, GLOW_v$ )
2:    $Q_n, Q_e, KG \leftarrow \text{entityExtraction}(Q)$  ▷ extract the question's node and edge
3:    $KG_{Sc} \leftarrow \text{getKGScheme}(KG)$  ▷ Load the KG schema triples
4:    $v_t, e_t \leftarrow \text{ER-Linking}(Q, KG_{Sc}, Q_n, Q_e)$  ▷ Link the question's node and edge crossponding KG's node and edge URIs
5:    $\mathcal{RC}_q \leftarrow \text{TextToSPARQL}(KG_{Sc}, v_t)$  ▷ generate  $\mathcal{RC}$  equivalent SPARQL query
6:    $\mathcal{RC}_{Triples} \leftarrow \text{execSPARQL}(\mathcal{RC}_q, KG)$  ▷ Execute the  $\mathcal{RC}$  SPARQL Query
7:    $L \leftarrow \text{getPossibleLabels}(v_t, e_t)$  ▷ Extract the set of possible labels
8:    $RC \leftarrow \text{RC-Serialization}(\mathcal{RC}_{Triples})$  ▷ Serialize the  $\mathcal{RC}$  triples into text.
9:    $GNN_{Ans} \leftarrow \text{GNNPredict}(v_t, e_t)$  ▷ predict the top-K GNN answers for  $v_t$  and  $e_t$ 
10:   $P \leftarrow \text{getPrompt}(Q, v_t, e_t, L, RC, GNN_{Ans}, GLOW_v)$  ▷ generate the GLOW's variation prompt
11:  return  $P$ 
12: end function

```

A.2 GLOW Prompt Examples

User Prompt:

Predict the chemical kingdom for the drug Yohimbine from the BioKG knowledge graph.
Answer:

Question Understanding Prompt:

<system>: You are an an expert Entity-Extraction NLP system.
 <user>: Given the following question, identify 1-The question main entity type, 2- the main entity, 3- the prediction label and 4- the KG name.
 Question: {}
 Answer:
 1-Question main entity: 2-Main Entity:
 3-Prediction label: 4-KG name:

Entity/Relation Linking Prompt:

<system>: You are an an expert knowledge graph entity-relation Linking NLP system.
 <user>: Given the following KG Schema in the basic graph pattern CSV format, one predicate per line: node type, relation, node type.

 KG Schema:
 {}

 1- What is the node type in the schema that corresponds to {main entity}? Return only the name.
 2- Choose from the schema the BGP (node type, relation, node type) that describes the {main entity type} value {main entity}. Return only the BGP.
 3- Choose from the schema the BGP (node type, relation, node type) that describe the {main entity type } {prediction label}. Return only the BGP.

 Answer:
 1-
 2-
 3-

Question To SPARQL Prompt:

<system>: You are an expert text-To-SPARQL translation system.

<user>: Given the following KG Schema in the basic graph pattern CSV format, one predicate per line: node type, relation, node type.

KG Schema:

{ }

Write a SPARQL query that selects the {main entity type} that satisfy the following BGPs.

1- {question node type} ,[label/name/title] ,{question node}

2- {Other BGPs}

graph prefix: {KG Prefix}

Answer: SPARQL Query

Do not return any explanation or reasoning details.

SPARQL Query Example:

PREFIX biokg: <http://www.biokg.com/>

SELECT ?drug as ?vt ?kingdom as ?vl

WHERE {

VALUES ?name { "Yohimbine" }

?drug biokg:NAME ?name .

?drug biokg:KINGDOM ?kingdom . }

Basic Prompt:

<system>: You are an expert open world question answer system.

<user>: Predict the {**Prediction Label**} for the following {**question node type**} from {**KG details**} knowledge graph.

- Do not return any context or analysis.

- Help: The possible list of {**Prediction Label Type**}s are: [{ **Labels List**}]

- Question: {**question node**}

Answer

To generate an instance of this template, replace the question node type, KG, and prediction label with values from one of the queries in table 5. **An example prompt for OWA-Q #1:**

<system>: You are an expert open world question answer system.

<user>: Predict the **Kingdom** for the following **Drug** from the **BioKG** , a **Biomedical** knowledge graph.

- Do not return any context or analysis.

- Help: The list of **Kingdoms** are: [Organic,Non-Organic]

- Question: **Durg**: Yohimbine

Answer:

GLOW-G Prompt:

<system>: You are an expert open world question answer system.

<user>: Predict the **Kingdom** for the following **Drug** from the **BioKG** , a **Biomedical** knowledge graph.

- Do not return any context or analysis.

- Help: The possible list of **Kingdoms** are: [Organic,Non-Organic]

- Question: **Durg**: and its associated triples. Capsaicin: [("DDI", "DB13677"), ..]

Answer:

GLOW-N Prompt:

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<system>: You are an expert open world question answer system.
<user>: Predict the Kingdom for the following Drug from the BioKG , a Biomedical knowledge graph.
- Do not return any context or analysis.
- Help: The possible list of Kingdoms are: [Organic,Non-Organic]
- Verify the following list of GNN Answers: [ Organic,Non-Organic, ..]
- Question: Durg: Yohimbine
Answer:
```

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GLOW-GN Prompt:

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```
<system>: You are an expert open world question answer system.
<user>: Predict the Kingdom for the following Drug from the BioKG , a Biomedical knowledge graph.
- Do not return any context or analysis.
- Help: The possible list of Kingdoms are : [Organic,Non-Organic]
- Verify the following GNN Answer: [ Organic]
- Question: Durg and its associated triples. Capsaicin: [("DDI", "DB13677"), ..]
Answer:
```

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A.3 LLM as-a-Judge Prompt

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<system>: You are an expert LLM-as-a-Judge system.
<user>: Given the following list of predicted and true pairs of values.
-Rank the predicted value against the true value using two metrics.
1- Exact Match Rule: you compare the two strings after normalization and remove any special characters. report 1 if both values are literally and semantically equal and 0 otherwise.
2- Hierarchical/Categorical Match Rule: report 1 if the predicted value is under a subcategory or hierarchically belongs to the true value or is a synonym and 0 otherwise.
- Example:
List of pairs: [[music, art], [painter, artist],[ football player, soccer player], [ lawyer, judge], [lawyer, player]]
Answer: [[0,1],[0,1],[1,1],[0,1],[0,0]]
- Question:
-List of pairs: {ListOfPairs}
-Note: refine each pair and return Answer for exactly {length(ListOfPairs)} pairs without explanation.
-Finally: make sure you return only {length(ListOfPairs)} pair of answers.
Answer:
```

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A.4 Full Details of Our GLOW-Bench

Table 5: Our GLOW-Bench benchmark characterizes each Open-World Question Template (OW-QT) across four KG-based features: (1) Knowledge Domain (KD): Domain-Specific (DS), Entertainment (E), or Generic (G); (2) Target Entity Type (ET); (3) Reasoning Hops (RH); and (4) Multiple Choice Count (MCC).

OW-QT	KG	KD	Target ET	RH	#Class	MCC	Label Type	Reasoning Path
1	BioKG	DS	Drug	1	2	2–4	Kingdom	Kingdom
2		DS	Drug	1	13	8–16	Superclass	Superclass
3		DS	Drug	2	29	16–32	Class	RelatedPubMed → Class
4		DS	Protein	1	18	16–32	SPECIES	SPECIES
5		DS	Protein	2	15	8–16	R.Keyword	RelatedPubMed → R.Keyword
6		DS	Protein	1	4	2–4	Family	Family
7	CrunchBase	DS	Investor	1	12	8–16	Country	Country
8		DS	Investor	1	14	8–16	InvestRegion	InvestRegion
9		DS	Investor	1	6	4–8	PositionTitle	PositionTitle
10		DS	Investor	1	11	8–16	Company	Company
11	LinkedMDB	E	Film	1	15	8–16	Language	Language
12		E	Film	1	27	16–32	Country	Country
13		E	Film	2	39	32+	Producer	Sequel → Producer
14		E	Film	2	7	4–8	Genre	Sequel → Genre
15		E	Film	1	28	16–32	Subject	Subject
16	YAGO4	G	CreativeW	2	13	8–16	ProdComp	byArtist → ProdComp
17		G	CreativeW	2	14	8–16	Publisher	isBasedOn → Publisher
18		G	CreativeW	1	25	16–32	PublishLang	Author
19		G	CreativeW	2	11	8–16	Genre	ProdComp → Genre
20		G	CreativeW	1	6	4–8	Country	Country
21		G	Person	1	3	2–4	GivenAward	GivenAward
22		G	Person	1	91	32+	Nationality	Nationality
23		G	Person	1	4	4–8	GraduateOfOrg	GraduateOfOrg
24		G	Person	1	102	32+	Occupation	Occupation
25		G	Person	1	8	4–8	SpokenLang	SpokenLang

A.5 Full Detailed Results

Table 6: The detailed accuracy (%) of GLOW compared to baseline models on existing question answering datasets from (Hu et al., 2024) and our developed benchmark GLOW-Bench. All results below use Qwen-3-8B as the underlying LLM. Best results are marked in **bold**.

LLM-Model	Dataset	RH	AskGNN	GraphSAINT	LLM-Only	GLOW-L	GLOW-G	GLOW-N	GLOW-GN
Qwen3:8b	Arxiv2023	1	62	41	31	59	64	66	81
	ogbn-arxiv	1	70	59	12	63	71	60	77
	ogbn-product	1	29	45	34	41	44	51	57
	drug-superclass	1	88	84	29	50	77	85	92
	drug-kingdom	1	98	96	88	91	98	97	96
	protein-SPECIES	1	75	89	8	19	71	91	92
	protein-FAMILY	1	67	70	9	35	39	69	71
	film-country	1	55	55	40	40	60	57	55
	film-subject	1	77	81	15	50	30	82	80
	film-language	1	87	83	70	73	80	85	86
	person-nationality	1	86	89	27	45	89	91	90
	parson-graduateOfOrg	1	51	53	3	27	36	54	55
	person-occupation	1	48	46	7	15	38	46	53
	person-spokenLang	1	86	83	70	83	89	83	85
	person-givenAward	1	93	91	7	75	93	93	93
	CWork-PublishedLang	1	75	80	50	55	55	85	95
	CWork-country	1	72	51	64	62	75	48	79
	person-title	1	79	72	5	63	47	77	83
	person-Company	1	97	95	3	30	21	96	97
	person-InvestmentRegion	1	98	94	7	95	87	96	100
	person-InvestmentCountry	1	98	95	38	94	55	97	97
	drug-class	2	46	68	11	33	45	70	73
	protein-keyword	2	35	42	5	33	53	44	57
	film-genre	2	54	51	50	60	62	51	52
	film-producer	2	35	21	7	14	24	21	28
	CWork-ProductionCompany	2	20	22	10	21	31	25	28
	CWork-Genere	2	20	22	16	22	33	19	27
	CWork-publisher	2	31	28	33	58	62	27	32

Table 7: The detailed accuracy (%) of GLOW compared to baseline models on existing question answering datasets from (Hu et al., 2024) and our developed benchmark GLOW-Bench. All results below use GPT-4O-Mini (OpenAI, 2024) as the underlying LLM. Best results are marked in **bold**.

LLM-Model	Dataset	RH	AskGNN	GraphSAINT	LLM-Only	GLOW-L	GLOW-G	GLOW-N	GLOW-GN
GPT-4o-Mini	Arxiv2023	1	N/A	41	35	58	59	48	62
	ogbn-arxiv	1	N/A	41	35	58	59	48	67
	ogbn-product	1	N/A	45	N/A	N/A	N/A	N/A	N/A
	drug-superclass	1	N/A	84	8	47	57	73	87
	drug-kingdom	1	N/A	96	31	100	100	98	99
	protein-SPECIES	1	N/A	89	6	10	77	92	93
	protein-FAMILY	1	N/A	70	11	53	43	69	72
	film-country	1	N/A	55	47	47	71	57	58
	film-subject	1	N/A	81	15	73	38	76	75
	film-language	1	N/A	83	80	80	80	86	88
	person-nationality	1	N/A	89	43	56	89	92	87
	parson-graduateOfOrg	1	N/A	53	19	22	33	55	57
	person-occupation	1	N/A	46	7	23	30	46	52
	person-spokenLang	1	N/A	83	70	81	91	83	89
	person-givenAward	1	N/A	91	6	68	96	93	91
	CWork-PublishedLanguage	1	N/A	80	65	60	65	83	87
	CWork-country	1	N/A	51	62	64	77	48	83
	person-title	1	N/A	72	5	38	55	74	77
	person-company	1	N/A	95	3	6	30	96	98
	person-InvestmentRegion	1	N/A	94	9	7	96	100	96
	person-InvestmentCountry	1	N/A	95	41	44	100	97	97
	drug-class	2	N/A	68	8	27	45	69	67
	protein-keyword	2	N/A	42	7	28	44	40	45
	film-genre	2	N/A	51	68	65	70	50	53
	film-producer	2	N/A	21	20	17	20	21	31
	CWork-ProductionCompany	2	N/A	22	10	28	42	25	32
	CWork-Genere	2	N/A	22	27	27	72	19	44
	CWork-publisher	2	N/A	28	62	70	79	25	75

Table 8: The detailed accuracy (%) of GLOW compared to baseline models on existing question answering datasets from (Hu et al., 2024) and our developed benchmark GLOW-Bench. All results below use DeepSeek-V3 (DeepSeek-AI, 2024) as the underlying LLM. Best results are marked in **bold**.

LLM-Model	Dataset	RH	AskGNN	GraphSAINT	LLM-Only	GLOW-L	GLOW-G	GLOW-N	GLOW-GN
DeepSeek-V3	Arxiv2023	1	N/A	41	17	50	57	61	73
	ogbn-arxiv	1	N/A	59	58	60	62	63	65
	ogbn-product	1	N/A	45	N/A	N/A	N/A	N/A	N/A
	drug-superclass	1	N/A	84	3	32	48	73	86
	drug-kingdom	1	N/A	96	9	88	100	98	96
	protein-SPECIES	1	N/A	89	2	8	79	92	93
	protein-FAMILY	1	N/A	70	21	76	87	69	81
	film-country	1	N/A	55	57	65	70	57	71
	film-subject	1	N/A	81	23	80	38	80	83
	film-language	1	N/A	83	81	86	86	86	93
	person-nationality	1	N/A	89	72	81	87	91	93
	parson-graduateOfOrg	1	N/A	53	32	48	43	52	54
	person-occupation	1	N/A	46	19	34	34	46	53
	person-spokenLang	1	N/A	83	79	79	82	83	85
	person-givenAward	1	N/A	91	3	56	58	93	94
	CWork-PublishedLanguage	1	N/A	80	47	50	50	84	90
	CWork-country	1	N/A	51	68	75	87	48	79
	person-title	1	N/A	72	41	41	63	74	77
	person-company	1	N/A	95	3	24	33	96	97
	person-InvestmentRegion	1	N/A	94	7	21	100	89	100
	person-InvestmentCountry	1	N/A	95	44	58	100	97	100
	drug-class	2	N/A	68	7	21	39	69	70
	protein-keyword	2	N/A	42	3	40	59	49	52
	film-genre	2	N/A	51	53	60	58	50	62
	film-producer	2	N/A	21	10	30	50	14	34
	CWork-ProductionCompany	2	N/A	22	28	53	57	25	53
	CWork-Genere	2	N/A	22	18	22	27	16	28
	CWork-publisher	2	N/A	28	70	79	87	16	70

Table 9: The detailed accuracy (%) of GLOW compared to baseline models on existing question answering datasets from (Hu et al., 2024) and our developed benchmark GLOW-Bench. All results below use DeepSeek-R1-7B (DeepSeek-AI, 2025) as the underlying LLM. Best results are marked in **bold**.

LLM-Model	Dataset	RH	AskGNN	GraphSAINT	LLM-Only	GLOW-L	GLOW-G	GLOW-N	GLOW-GN
DeepSeek-r1:7B	Arxiv2023	1	48	41	24	44	55	59	67
	ogbn-arxiv	1	50	59	13	30	30	61	63
	ogbn-product	1	19	45	27	36	38	43	51
	drug-superclass	1	53	84	20	20	21	69	68
	drug-kingdom	1	98	96	60	90	100	98	100
	protein-SPECIES	1	71	89	1	4	62	88	90
	protein-FAMILY	1	62	70	7	30	35	68	69
	film-country	1	56	55	17	44	52	57	64
	film-subject	1	78	81	8	11	34	76	82
	film-language	1	86	83	73	73	66	86	88
	person-nationality	1	85	89	22	30	79	91	89
	parson-graduateOfOrg	1	50	53	5	19	36	53	54
	person-occupation	1	47	46	3	15	31	46	54
	person-spokenLang	1	82	83	35	41	79	81	83
	person-givenAward	1	89	91	8	50	70	90	92
	CWork-PublishedLanguage	1	71	80	50	50	55	77	81
	CWork-country	1	62	51	37	41	70	48	66
	person-title	1	73	72	5	44	52	80	82
	person-company	1	93	95	3	30	19	96	92
	person-InvestmentRegion	1	92	94	3	64	96	100	100
	person-InvestmentCountry	1	93	95	11	72	94	97	97
	drug-class	2	41	68	4	15	24	69	71
	protein-keyword	2	31	42	3	18	22	32	38
	film-genre	2	53	51	35	50	56	50	57
	film-producer	2	30	21	7	7	7	14	24
	CWork-ProductionCompany	2	18	22	10	17	21	25	32
	CWork-Genere	2	17	22	11	11	16	16	27
	CWork-publisher	2	25	28	12	20	40	16	26

Table 10: The detailed accuracy (%) of GLOW compared to baseline models on existing question answering datasets from (Hu et al., 2024) and our developed benchmark GLOW-Bench. All results below use IBM Granite3.3:8B (Granite Team, 2025) as the underlying LLM. Best results are marked in **bold**.

LLM-Model	Dataset	RH	AskGNN	GraphSAINT	LLM-Only	GLOW-G	GLOW-N	GLOW-GN
Granite3.3:8B	Arxiv2023	1	63	41	14	32	63	77
	ogbn-arxiv	1	66	59	5	31	55	73
	ogbn-product	1	30	45	22	45	52	54
	drug-superclass	1	86	84	0	21.79	73.08	70.51
	drug-kingdom	1	97	96	7.81	95.31	98.44	98.44
	protein-SPECIES	1	75.14	89	2.41	44.58	92.77	90.36
	protein-FAMILY	1	67.23	70	0	35.9	69.23	71.79
	film-country	1	55.67	55.5	47.5	65.79	57.89	55.26
	film-subject	1	76.4	81	11.54	26.92	80.77	84.62
	film-language	1	84.15	83	86.67	80	86.67	87.67
	person-nationality	1	85.6	89.1	52.08	79.17	90.67	91.67
	parson-graduateOfOrg	1	52.12	53.2	5.56	30.56	55.56	56.78
	person-occupation	1	46.5	46.15	7.69	23.08	46.15	50
	person-spokenLang	1	85.4	83.33	48.33	52.08	83.33	88.33
	person-givenAward	1	92.8	91.7	3.13	62.5	93.75	94.38
	CWork-PublishedLanguage	1	81	80	45	80	81	94
	CWork-country	1	71.8	51	58.33	75	58.89	68.75
	person-title	1	78.4	72.44	13.89	63.89	80	83.33
	person-company	1	96.6	95.8	3.03	42.42	95.75	96.97
	person-InvestmentRegion	1	99	94.8	3.57	82.14	98	100
	person-InvestmentCountry	1	98.8	95.5	38.89	77.78	96.14	97.22
	drug-class	2	47	68	0	27.27	69.7	69.7
	protein-keyword	2	35.17	42	1.49	36.46	9.38	37.96
	film-genre	2	54.3	51	35	51.67	54.17	56.33
	film-producer	2	35	21	5	30	23	29
	CWork-ProductionCompany	2	21.8	22	7.14	21.43	25	32.14
	CWork-Genre	2	21.33	22.45	16.67	50	16.67	33.33
	CWork-publisher	2	32.7	28.2	33.33	58	26.67	36.47